

PEAD Strategies Overview

One at-a-glance catalog of every strategy design in this project: what it does, how it performed, and whether it needs a GPU.

Companion to *PEAD_Strategy_Showcase.pdf*, *PEAD_Strategy_Development.pdf*, and *PEAD_Options_Strategy_Report.pdf* -- this report is the index, not the deep-dive.

This project has built **9 equity strategy designs** (Strategies 1-8, with Strategy 6 also having a beta-overlay variant) and explored **one options strategy across 4 GPU-tiered stages** -- Tier 1 through Tier 4 are not 4 separate, independent options strategies; they are one lineage, each tier built only because the tier before it left a concrete question unanswered. Of those 5 options rows, only Tiers 1, 1.5, and 4 currently have a complete, real backtested Sharpe/return/drawdown; Tiers 2 and 3 are still parameter-search/screening stages -- their status column says so explicitly rather than showing a number that doesn't exist yet.

How to read "GPU needed?" Every equity strategy 1-7 and options Tier 1/1.5 runs fine on a laptop CPU in seconds. GPU only matters for the compute-heavy tiers built for the dedicated 5090 machine: Strategy 8's walk-forward model training and parameter sweep, and options Tiers 2-4's large batched tensor searches.

Equity strategies (9 designs)

Strategy	What it does	Ann. Return	Ann. Vol	Sharpe	Max DD	GPU?
1. Extreme Decile L/S	Long the most positive earnings-surprise decile, short the most negative, holding each decile/size/book-to-market cell for its own empirically-best number of days.	4.91%	5.32%	0.93	-18.48%	No (CPU, seconds)
2. Rank-Weighted	Trades every SUE decile at once, weighted continuously by rank distance from the median for broader diversification.	3.88%	3.86%	1.01	-13.96%	No (CPU, seconds)
3. Balanced	The extreme-decile trade, neutralized by size quintile each quarter so no single size bucket dominates the book.	4.74%	5.26%	0.91	-17.53%	No (CPU, seconds)
4. Trimmed/Tilted	Drops the low-conviction middle of the SUE distribution, tilts weight toward the extremes, and trims lowest-conviction positions first when a cap binds.	5.26%	4.41%	1.18	-13.07%	No (CPU, seconds)
5. + Sector-Neutral	Strategy 4 plus Fama-French 12-sector neutrality, fixing a hidden sector-concentration risk -- best Sharpe and shallowest drawdown of the hand-built designs.	4.06%	3.02%	1.33	-10.83%	No (CPU, seconds)
6. + Leverage Cap 2.5x	Strategy 5 with its leverage cap raised from 1.5x to 2.5x, since diagnostics showed nothing left for the tighter cap to protect against.	6.49%	4.94%	1.30	-16.63%	No (CPU, seconds)
6+Beta. Beta Overlay	Strategy 6's untouched alpha book plus a separate 0.5x-NAV market-index position held alongside it, pushing return toward equity-like levels.	10.08%	10.51%	0.97	-23.23%	No (CPU, seconds)
7. Unconstrained Net Exposure	Strategy 5/6's signal without forcing long and short dollar totals to match each quarter -- a documented cautionary result, not a recommendation.	7.30%	6.56%	1.11	-15.54%	No (CPU, seconds)
8. GPU/ML-Tiered	Blends SUE rank with a walk-forward-trained neural signal built on trailing momentum/volatility features, with sizing and leverage chosen by a real grid search instead of hand-picked constants.	3.51%	3.77%	0.93	-12.93%	Yes (walk-forward training + parameter sweep are GPU-batched -- 12.3x faster on a Colab T4 than this dev CPU)

Table 1. Figures 1996-2013, market-adjusted (raw for Strategy 7) daily returns, trailing-NAV compounding, \$10,000,000 starting capital. Recommended for live consideration: Strategy 6 (pure alpha) or Strategy 6+Beta (higher return, more market-correlated risk).

Strategy 8 status: Real run, full sample: the GPU param sweep itself picked `ml_blend_weight=0.0` (pure SUE rank beat every ML-blended variant) and holdout Sharpe (0.42) trailed search-window Sharpe (1.14) -- the ML signal does not add real value in this data and Strategy 8 underperforms Strategies 5/6. Kept as a documented result, like the EAR and SUE-reversal dead ends. Walk-forward mean out-of-sample IC across all 15 freshly-trained yearly folds was 0.0095 -- essentially no signal. Confirmed on a real Colab T4 GPU as well as this CPU dev machine (identical numbers on both, 12.3x faster on the T4: 660s -> 55s for the walk-forward training step alone).

Strategy Comparison: Return, Risk-Adjusted Return, and Drawdown

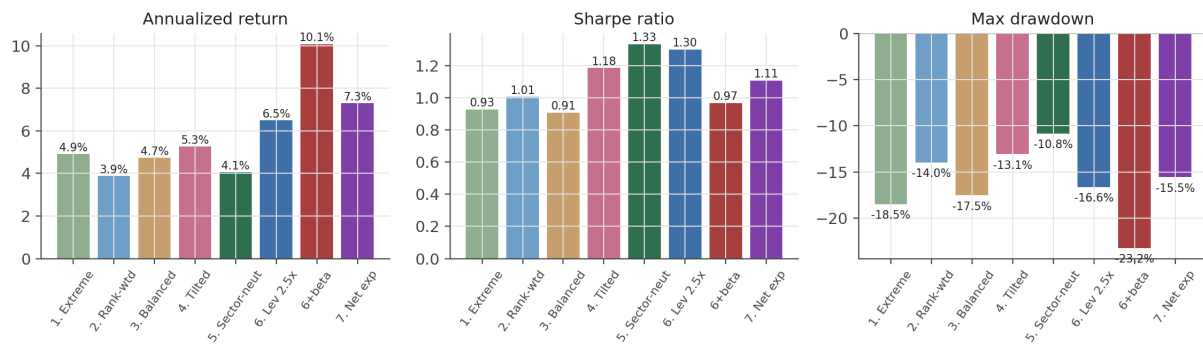


Figure 1. Return/Sharpe/drawdown comparison across all 8 hand-built equity strategies.

Options strategy (1 lineage, 5 tiers)

Tier	What it does	Ann. Return	Ann. Vol	Sharpe	Max DD	GPU?
Tier 1. Near-ATM Heuristic	Buys a near-50-delta call (long-SUE deciles) or put (short-SUE deciles) at earnings and holds to a fixed horizon, with real capital sizing and trading costs.	9.51%	9.16%	1.04	-25.90%	No (CPU, seconds)
Tier 1.5. Kelly-Optimal Selector	Scores every strike in the real day-0 chain by expected log-growth against an empirical, walk-forward return distribution, sized at half-Kelly.	15.31%	4.03%	3.56	-21.36%	Optional (ran on CPU; GPU-batchable at full-chain scale)
Tier 2. Dynamic Exit Optimizer	Searches for the best stop-loss/profit-target exit rule per decile bucket instead of a fixed holding horizon.	N/A	N/A	N/A	N/A	Yes (dense position x day x parameter-combo tensor scan)
Tier 3. Sizing Sweep + ML Selector	A GPU grid search over trim/tilt/horizon to screen sizing constants, plus a planned Optuna/PyTorch model testing IV/liquidity/sector features against decile membership.	N/A	N/A	N/A	N/A	Yes (GPU-batched sweep; Optuna + PyTorch training)
Tier 4. Joint Equity+Options +Beta Allocation	Searches the best combined weighting across the equity alpha book, the options sleeve, and the market-beta overlay held together.	14.56%	7.64%	1.82	-16.63%	Yes (largest tensor sweep in the project)

Table 2. Figures at the 60-day horizon, 1996-2013 ex-2011, real capital sizing and trading costs, \$10,000,000 starting capital. Tier 1.5 is the currently-recommended options design.

Tier 2. Dynamic Exit Optimizer status: Parameter search only, run at small/mock scale so far -- not yet folded into a full capital-sized backtest.

Tier 3. Sizing Sweep + ML Selector status: Sizing sweep completed on real full-scale data as a screening pass (not a final Sharpe); the ML selector has not been run.

Tier 4 leverage caveat: the best combo found ($w_{\text{equity}}=2.0$, $w_{\text{options}}=1.0$, $w_{\text{beta}}=0.25$) assumes weights above 1.0x are literally free, unlimited leverage on top of Strategy 6's already-2.5x internal leverage. Treat the 1.82 Sharpe / 14.56% return as an upper-bound research signal about where blending helps, not an achievable number, until a real cost-of-leverage term is added.

Tier 1 vs. Tier 1.5, by hold horizon

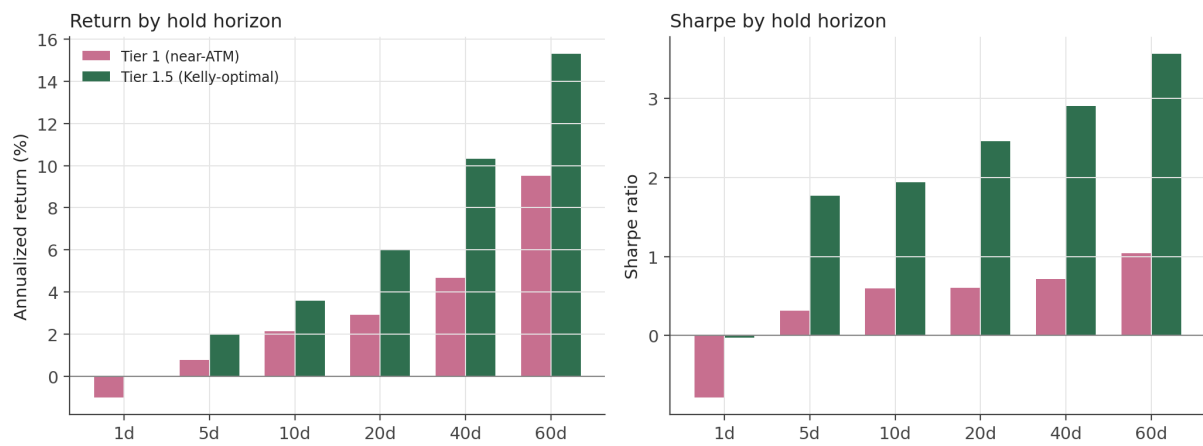


Figure 2. Tier 1 (near-ATM heuristic) vs. Tier 1.5 (Kelly-optimal) by holding horizon.